

GCS Holdings

Outperform

Initiation: Riding the AI optical upgrade cycle

Key Points

- GCS is a key optical chip supplier, positioned to benefit from an 89% CAGR we forecast for the transceiver industry over 2025-28E.
- We expect PD spec upgrades to lift ASPs 6-8x and expand gross margins, supported by a US\$90mn+ backlog, equivalent to more than one year of revenue.
- The recent substrate shortage and AI-driven sell-off have weighed on GCS's shares, creating what we believe is an attractive risk-reward opportunity.

- GCS is a leading Indium Phosphide (InP) Photodetector (PD) manufacturer.** PDs are used in optical transceivers to convert optical signals into electrical data. The AI revolution is driving demand for GPU and the AI networks and optical infrastructure that connects them. We forecast an 89% CAGR for the transceiver market over 2025-28E. We estimate GCS held 70% of the 100G transceiver PD market in 2025. With a PD backlog exceeding US\$90mn, equivalent to more than one year of revenue, we expect GCS to grow alongside the expanding market.
- 1.6T transceiver taking off in 2026, bringing significant PD content growth:** As AI clusters scale and network speeds migrate from 800G to 1.6T starting 2H26 (see Fig 16), we expect a 6-8x ASP increase in 200 PD, and further GPM upside for GCS.
- Overflow demand benefit both PD and laser.** Industry InP capacity is tight. Additionally, beyond PDs, GCS is expanding into CW lasers, creating a potential second growth engine from 2027. With demand outpacing supply, rising technology content, and limited industry capacity, we believe GCS is emerging as one of the most compelling yet underappreciated beneficiaries of the AI optical networking cycle.

Valuation: Our NT\$710 target price is based on 32x 2028E P/E, supported by a 87% 2026-28E EPS CAGR, driven by the company's strengthening position as a leading photodetector (PD) supplier and its growing exposure to the laser market.

Catalysts: 200G PD volume ramp and customer qualification, conversion of the US\$90mn+ backlog into revenue, CW laser qualification.

Investment Thesis and Recommendation

We believe the market underestimates the earnings contribution from 200G PD adoption, margin expansion from higher-value optical products and potential for GCS to benefit from constrained industry InP capacity. As these drivers become more evident, we expect GCS to re-rate, supporting our O/F call.

Semicons & Semicon Equip Taiwan



Nick Hung



Leo Tsai

4991 TT	Outperform
Price (at 26 Aug 2026)	TWD445.00
12-month target	TWD710.00
12-month TSR (%)	60.4
Volatility Index	Very High
Market Cap (Local) (m)	54,003
Market Cap (USD) (m)	1,694
Free Float (%)	65
30-day avg turnover (USD) (m)	55.1

Investment Fundamentals

Year end 31 Dec	2025A	2026E	2027E	2028E
Revenue (m)	2,195	4,151	8,151	12,711
Revenue growth (%)	25.4	89.1	96.4	55.9
EBIT (m)	403.9	814.5	1,910	3,471
EBIT growth (%)	109.5	101.7	134.5	81.7
Reported profit (m)	17.5	758.2	1,518	2,657
Adjusted profit (m)	17.5	758.2	1,518	2,657
EPS rep [TWD]	0.2	6.4	12.7	22.2
EPS rep growth (%)	NM	4,044.6	99.4	75.0
EPS adj [TWD]	0.2	6.4	12.7	22.2
EPS adj growth (%)	NM	3,957.4	99.3	75.0
Net debt/equity (%)	(30.3)	(14.6)	(13.2)	(14.1)
ROA (%)	11.0	18.4	34.4	48.2
ROE (%)	0.6	19.9	31.1	40.9
PER rep (x)	2,899.9	70.0	35.1	20.0
PER adj (x)	2,838.3	70.0	35.1	20.0
EV/EBITDA (x)	298.9	53.3	22.9	13.3
P/BV (x)	15.5	12.2	9.9	7.0
Total div yield (%)	0.9	0.9	0.9	0.9
Quant (rank vs. global sector)	570 / 670			

4991 TT rel TAIEX performance, & rec history



Source: FactSet, Macquarie Research, Aug 2026 (all figures in TWD unless noted, TP in TWD)

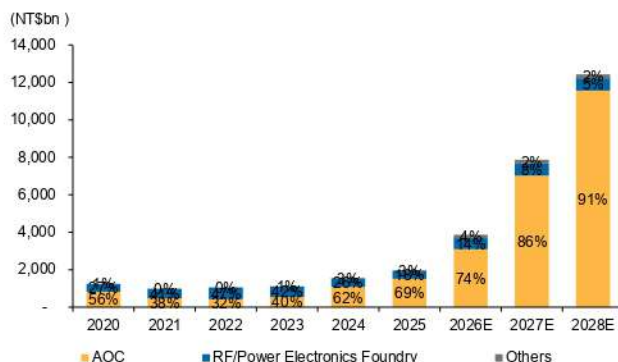
Key Risks to Investment Thesis

- Delayed 1.6T transceiver shipment.
- Delayed 200G PD development.
- Failure to qualified CW laser.
- Quicker timeline of SiPh / CPO adoption.

Company Description

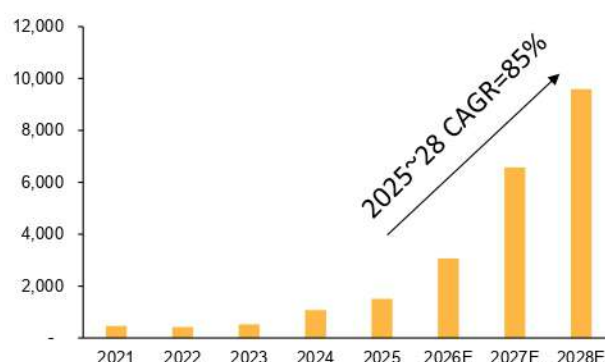
GCS is a leading manufacturer of Indium Phosphide (InP) chips used in optical transceivers. Its core product, photodetectors (PDs), converts optical signals into electrical data and is a critical component in high-speed AI and data centre networks. The company holds an estimated 70% share of the 100G PD market.

Figure 1 - Revenue breakdown



Source: Company data, Macquarie Research, Aug 2026

Figure 2 - GCS PD revenue



Source: Company, Macquarie Research, Aug 2026

GCS Holdings (4991 TT) TWD/(m) unless otherwise noted

Income Statement	Dec FY	2025A	2026E	2027E	2028E	Q2/26A	Q3/26E	Balance Sheet	2025A	2026E	2027E	2028E
Revenue		2,195	4,151	8,151	12,711	770.1	1,110	Cash	1,210	874.3	948.3	1,306
Cost of Goods Sold		1,161	2,174	4,081	5,935	436.9	594.6	Receivables	364.7	824.1	1,034	1,879
Gross Profit		1,035	1,977	4,070	6,776	333.2	515.7	Inventories	457.9	1,161	1,674	2,480
EBITDA		534.8	1,052	2,254	3,874	153.9	288.7	Investments	527.6	254.7	254.7	254.7
Depreciation		128.5	234.5	340.6	400.6	55.1	72.0	Fixed Assets	754.5	975.5	1,235	1,434
Amortisation		2.4	3.1	3.1	3.1	0.8	0.8	Intangibles	0.0	0.0	0.0	0.0
EBIT		403.9	814.5	1,910	3,471	98.0	215.9	Other Assets	530.9	926.6	943.5	960.4
Net Interest Income		9.7	27.5	27.5	27.5	6.9	6.9	Total Assets	3,846	5,017	6,089	8,316
Associates		(372.0)	(81.9)	0.0	0.0	0.0	0.0	Payables	27.5	69.8	100.6	149.1
Exceptionals		0.0	0.0	0.0	0.0	0.0	0.0	Short Term Debt	76.6	85.8	85.8	85.8
Forex Gains / Losses		0.5	(17.5)	0.0	0.0	0.0	0.0	Long Term Debt	150.4	151.3	151.3	151.3
Other Pre-Tax Income		(28.6)	242.2	114.6	114.6	28.7	28.7	Other Liabilities	347.0	346.6	346.6	346.6
Pre-Tax Profit		13.4	984.8	2,052	3,613	133.6	251.5	Total Liabilities	601.4	653.5	684.3	732.8
Tax Expense		4.1	(242.2)	(554.1)	(975.5)	(29.0)	(67.9)	Shareholders' Funds	3,169	4,438	5,479	7,658
Net Profit		17.5	742.6	1,498	2,637	104.5	183.6	Minority Interests	0.0	0.0	0.0	0.0
Minority Interests		0.0	15.6	20.0	20.0	5.6	5.0	Other	75.3	(74.8)	(74.8)	(74.8)
Reported Earnings		17.5	758.2	1,518	2,657	110.2	188.6	Total S/H Equity	3,244	4,363	5,405	7,583
Adjusted Earnings		17.5	758.2	1,518	2,657	110.2	188.6	Total Liab & S/H Funds	3,846	5,017	6,089	8,316
Basic Shares Outstanding		112.8	119.7	119.7	119.7	119.7	119.7	Net Debt / Equity (%)	(30.3)	(14.6)	(13.2)	(14.1)
Diluted Shares Outstanding		113.8	119.2	119.7	119.7	119.7	119.7	Interest Cover (x)	NM	NM	NM	NM
EPS (rep) [TWD]		0.2	6.4	12.7	22.2	0.9	1.6	ROE (%)	0.6	19.9	31.1	40.9
EPS (adj) [TWD]		0.2	6.4	12.7	22.2	0.9	1.6	ROA (%)	11.0	18.4	34.4	48.2
EPS adj (€)		0.2	6.4	12.7	22.2	0.9	1.6	ROIC (%)	20.1	27.2	37.4	54.0
Total DPS [TWD]		3.9	4.0	4.0	4.0	0.0	4.0					
Total DPS (€)		3.9	4.0	4.0	4.0	0.0	4.0					
Ratio		2025A	2026E	2027E	2028E			Cash Flow Analysis	2025A	2026E	2027E	2028E
Revenue Growth (%)		25.4	89.1	96.4	55.9	-	-	EBITDA	534.8	1,052	2,254	3,874
EBITDA Growth (%)		58.1	96.7	114.2	71.9	-	-	Tax Paid	4.1	(242.2)	(554.1)	(975.5)
EBIT Growth (%)		109.5	101.7	134.5	81.7	-	-	Chgs in Working Cap	(105.1)	(934.3)	(691.0)	(1,604)
EPS Growth (adj) (%)		NM	3,957.4	99.3	75.0	-	-	Net Interest Paid	9.7	27.5	27.5	27.5
Gross Profit Margin (%)		47.1	47.6	49.9	53.3	-	-	Other	(47.4)	193.8	114.6	114.6
EBITDA Margin (%)		24.4	25.3	27.6	30.5	-	-	Operating Cashflow	396.1	96.9	1,151	1,437
EBIT Margin (%)		18.4	19.6	23.4	27.3	-	-	Acquisitions	0.0	0.0	0.0	0.0
Net Profit Margin (%)		0.8	18.3	18.6	20.9	-	-	Capex	(59.4)	(490.8)	(600.0)	(600.0)
Payout Ratio (%)		2,518.3	62.9	31.5	18.0	-	-	Other	206.0	(311.6)	0.0	0.0
PE (rep) (x)		2,899.9	70.0	35.1	20.0	-	-	Investing Cashflow	146.6	(802.4)	(600.0)	(600.0)
PE (adj) (x)		2,838.3	70.0	35.1	20.0	-	-	Dividend (Ordinary)	0.0	(462.8)	(476.8)	(478.9)
EV/EBITDA (x)		298.9	53.3	22.9	13.3	-	-	Equity Raised	0.0	786.0	0.0	0.0
EV/EBIT (x)		1,528.6	70.6	27.1	14.9	-	-	Debt Movements	56.1	8.2	0.0	0.0
Price/Book (x)		15.5	12.2	9.9	7.0	-	-	Other	51.0	3.4	0.0	0.0
Total Div Yield (%)		0.9	0.9	0.9	0.9	-	-	Financing Cashflow	107.1	334.8	(476.8)	(478.9)
								Net Chg in Cash/Debt	633.9	(336.0)	74.0	358.1
								Free Cashflow	336.7	(393.9)	550.8	837.0

Source: Company data, Macquarie Research Aug 2026

Investment thesis

- **Macquarie's Asia Tech research team has highlighted exponential growth in AI networking demand.** Following our AI networking [report](#), we continue to forecast strong demand driven by larger AI clusters, increasingly hybrid network architectures and Agentic AI. We forecast a 57% CAGR for the networking switch market and an 89% CAGR for the transceiver market over 2025-28E. As AI clusters scale, additional layers of interconnect should drive significantly higher demand for networking switches and transceivers.
- **GCS is a III-V semiconductor manufacturer specialising in InP technologies.** Positioned upstream in the InP optical chip value chain, it provides wafer fabrication services to customers in optical networking, data centres, telecoms and consumer applications. GCS shipped more than 170mn 100G PD dies in 2025, implying a 65-70% share of the 800G PD market. The company has 25 years of compound semiconductor manufacturing experience, deep process expertise and long-standing customer qualifications. Complex materials engineering and high technical barriers limit new entrants, while lengthy qualification processes and long product lifecycles support recurring customer orders.
- **Strong demand continues to outstrip supply, giving GCS strong revenue visibility and opportunities to expand capacity.** Management highlighted a US\$92mn backlog, exceeding 2025 full-year revenue and supporting sustained demand. We expect 100G PD revenue to more than double from NT\$1.1bn in 2025 to NT\$2.2bn in 2026, driven by strong AI data-centre and optical networking demand.
- **Beyond its leading 100G PD position, GCS is preparing for the next upgrade cycle with 200G PDs** for 1.6T optical modules and is targeting a 25% market share following mass production in 2Q26. The company is exiting lower-margin RF and foundry businesses to free capacity for higher-growth InP-based optical products. We expect this transition to increase output, while CW laser commercialisation and targeted capex support GCS's ability to capture growing demand across the AI optical ecosystem.
- **Strategy focus eliminates China JV earnings drag.** Management expects to complete the China joint venture transaction by end-March 2026, reducing its ownership stake to 12%. We expect the transaction to remove a key source of non-operating earnings drag, including the US\$12mn investment loss recorded in 2025, and reduce the impact of future JV losses on reported earnings.

Valuation

- We initiate coverage of GCS with an Outperform rating and NT\$710 target price, based on 32x 2028E EPS. We forecast an 87% EPS CAGR over 2026-28E, driven by GCS's leading position in photodetectors, exposure to the 100G-to-200G PD upgrade cycle and growing participation in the laser market. We believe the market underestimates the earnings impact of 200G PD adoption, product-mix improvement and constrained industry InP capacity, which support a multi-year earnings inflection.
- Our 32x target multiple is supported by GCS's strong earnings growth, improving RoE and expanding exposure to a rapidly growing AI optical networking market. While we apply a discount to the peer average multiple of 35x, reflecting PDs' lower ASP relative to lasers, we believe this still supports attractive upside as the market recognises GCS's strengthening competitive position and earnings trajectory.
- We believe the recent share-price weakness does not reflect these long-term growth drivers. The fundamental outlook for AI optical networking remains intact, while improving substrate availability should help support future growth.

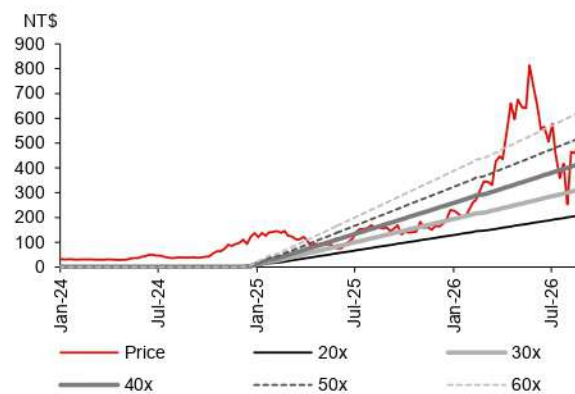
Figure 3 - Valuation comparisons

Ticker	Company	Rating	Mkt Cap (US\$m)	CP (LC)	TP (LC)	TSR (%)	EPS (LC)			PER (X)			P/BV (X)			ROE (%)			Div yield (%)		
							25A	26E	27E	25A	26E	27E	25A	26E	27E	25A	26E	27E	25A	26E	27E
InP substrate/LD/PD/Epi																					
2455.TT	VPEC	O	2,214	406	600	50%	3.0	5.6	11.0	137.1	73.1	36.9	22.8	18.7	14.1	16.5	28.1	43.6	0.6	1.0	1.9
3081.TT	Landmark	O	8,830	2,920	3,345	15%	3.9	18.8	46.9	757.9	155.1	62.3	77.5	52.3	30.7	10.8	41.0	62.1	0.1	0.3	0.6
AAOI.US	AAOI	NR	9,102	108	NA	NA	-0.3	0.7	4.6	na.	158.3	23.4	11.0	5.5	3.9	-7.9	28.0	18.0	0.0	0.0	na
COHR.US	Coherent	NR	53,950	275	NA	NA	4.4	7.4	11.9	63.2	37.4	23.2	6.0	4.8	4.1	3.9	12.4	20.0	0.0	0.0	0.0
LITE.US	Lumentum	NR	74,466	830	NA	NA	4.2	14.8	28.0	196.7	56.0	29.6	63.1	14.6	10.0	-118.8	-103.3	35.9	0.0	0.0	0.0
5802.JP	Sumitomo Ele	NR	43,821	2,200	NA	NA	82.5	138.9	127.9	26.7	15.8	17.2	2.8	2.4	2.2	13.2	13.1	13.7	1.5	1.9	2.2
5801.JP	Furukawa	NR	17,732	4,000	NA	NA	74.6	147.4	167.8	53.6	27.1	23.8	7.5	6.0	4.9	16.8	21.0	22.8	0.8	0.6	0.7
688498.CH	Yuanjie	NR	28,755	1,553	NA	NA	1.5	6.1	10.9	1,005.3	253.8	142.4	82.7	64.3	45.4	8.7	27.3	36.7	0.0	0.1	0.1
3105.TT	Win Semi	NR	5,040	379	NA	NA	4.0	7.3	10.2	94.8	52.3	37.2	3.9	3.8	3.5	4.3	7.5	10.0	0.5	0.8	1.8
AXTI.US	AXT	NR	4,288	65	NA	NA	-0.4	0.9	1.9	na.	75.6	34.0	11.1	na	na	-9.3	9.4	15.7	0.0	na	na
5016.JP	JK	NR	22,746	3,820	NA	NA	109.9	165.3	201.2	34.8	23.1	19.0	5.3	4.7	3.7	14.4	20.4	21.3	1.3	1.2	1.2
002428.CH	Yunnan Germ	NR	9,153	94	NA	NA	0.0	0.5	1.4	3,141.0	203.5	69.1	42.2	35.7	23.5	1.4	19.7	42.6	0.0	0.2	0.5
Average PER										551.1	94.3	35.2									
4991.TT	GCS	O	1,693	435	710	64%	0.2	6.4	12.7	2,774.5	68.4	34.3	15.1	11.9	9.6	0.6	19.9	31.1	0.9	0.9	0.9

Source: Bloomberg, Macquarie Research, August 2026

Closing price as of 25 August 2026

Figure 4 - 12-month forward P/E band



Source: TEJ, Macquarie Research, Aug 2026

Figure 5 - GCS EPS vs RoE



Source: Company, Macquarie Research, Aug 2026

Figure 6 - Sales Revenue and YoY



Source: Company data, Macquarie Research, Aug 2026

Figure 7 - GPM/OPM trend

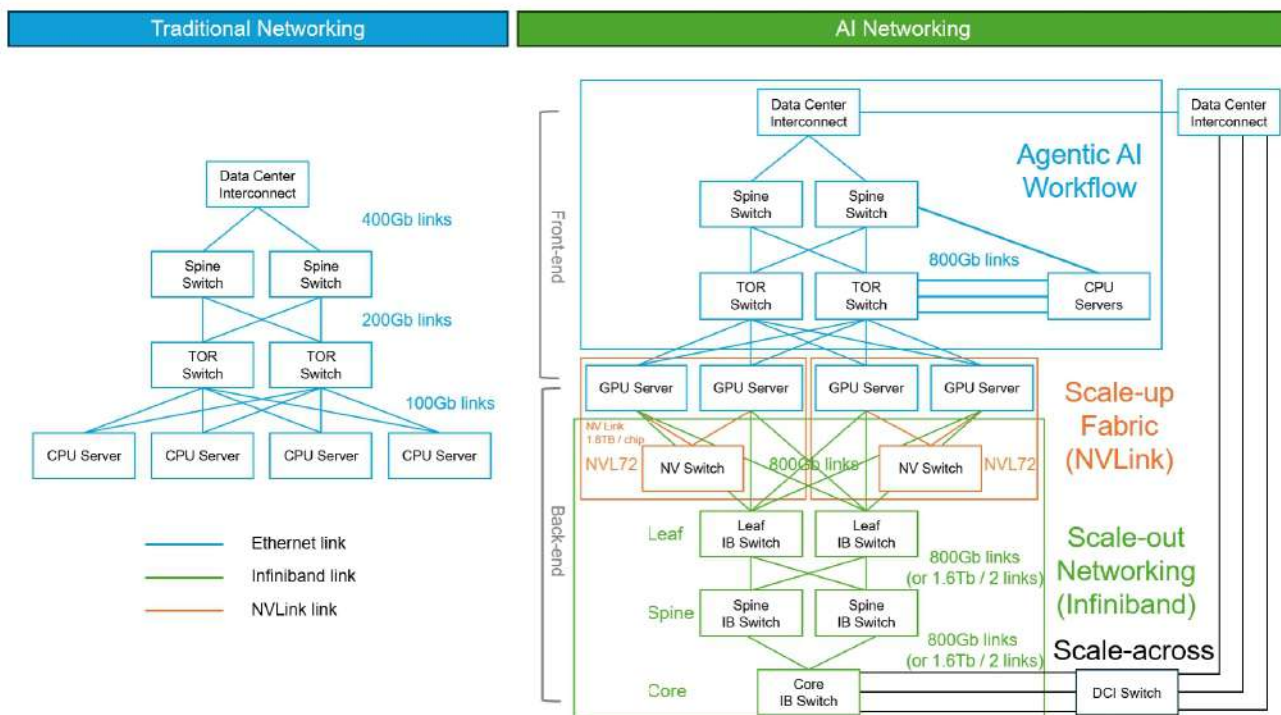


Source: Company data, Macquarie Research, Aug 2026

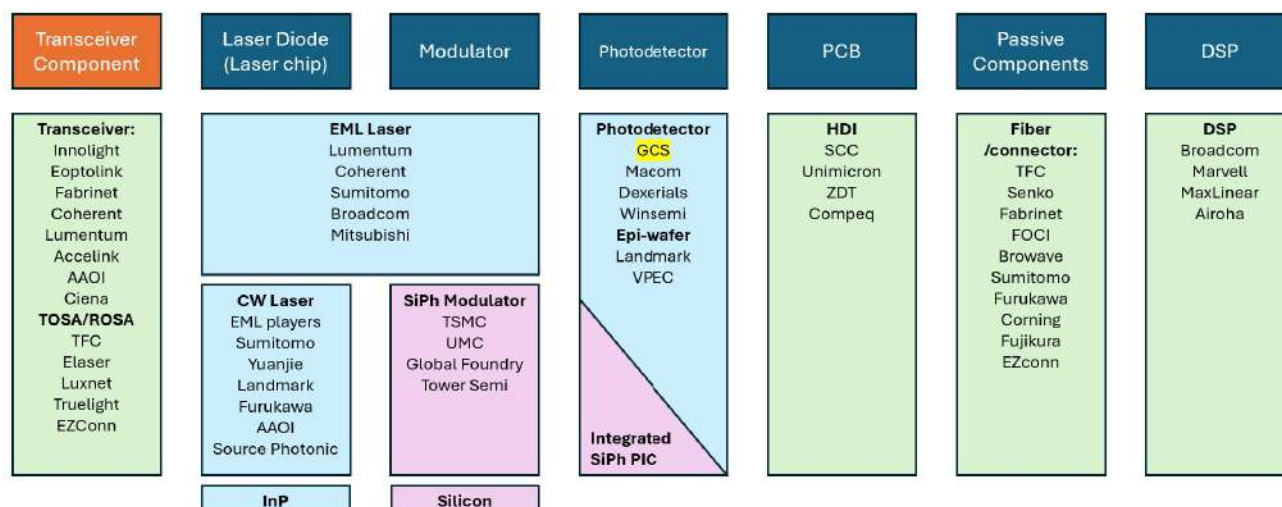
Transceivers set to expand through 2026-28E

- As highlighted in our AI networking thematic [report](#), rising data-transmission demand from AI training and Agentic AI should drive an 89% CAGR in the transceiver market over 2025-28E.
- We believe the sector is experiencing a cyclical upturn supported by secular AI demand growth. This demand has created supply-chain bottlenecks, and we expect shortages in key components, including InP substrates, EML lasers and DSPs, to persist through 2H26, with supply-demand conditions remaining tight in 2027. Transceivers require both laser diodes (LDs) and photodetectors (PDs).
- GCS supplies PDs and held a 70% share of the 100G PD market in 2025. We forecast 109% HoH growth in AOC revenue in 2H26E and 104% YoY growth in 2026E. We expect GCS to benefit from rising demand for 100G and 200G PDs as industry capacity shifts toward higher-ASP, higher-margin laser products, tightening PD supply.

Figure 8 - AI networking architecture change is driving 800G+ transceiver demand



Source: Macquarie Research, Aug 2026

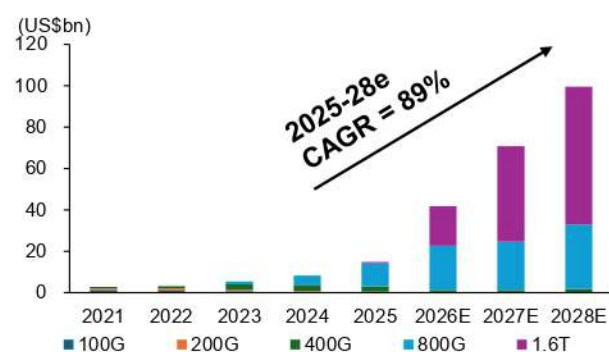
Figure 9 - Transceiver supply chain

Source: Macquarie Research, Aug 2026

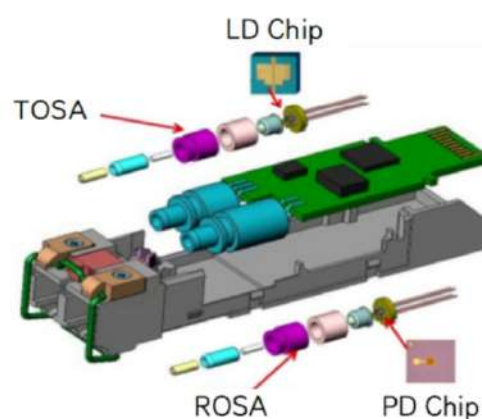
Figure 10 - MQ transceiver shipment (incl. optical engine) forecast

(mn units)	2021	2022	2023	2024	2025	2026E	2027E	2028E
100G	10.0	8.5	7.1	4.1	2.3	1.8	1.4	1.2
200G	3.1	4.8	4.0	2.8	2.5	2.2	2.0	1.5
400G	1.6	3.3	8.9	13.7	12.3	4.8	5.9	9.1
800G	0.0	0.0	1.7	7.3	20.2	46.5	59.7	79.8
1.6T	0.0	0.0	0.0	0.0	0.6	20.8	63.2	91.3

Source: Macquarie Research, Aug 2026

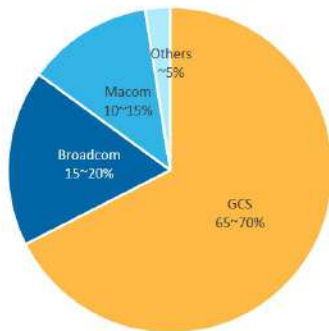
Figure 11 - MQ transceiver market forecast

Source: Macquarie Research, Aug 2026

Figure 12 - Transceiver component overview

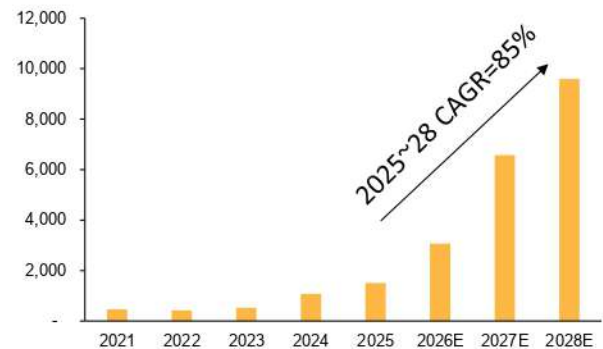
Source: Macquarie Research, Aug 2026

Figure 13 - Global 100G PD market share



Source: Macquarie Research, Aug 2026

Figure 14 - GCS PD revenue



Source: Company, Macquarie Research, Aug 2026

Optical upgrade cycle drives ASP upside

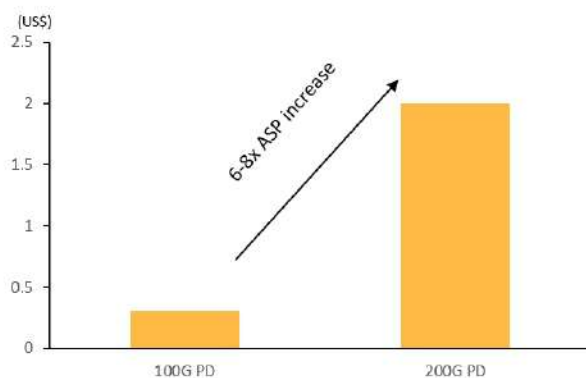
- Strong AI networking demand is accelerating the upgrade from 800G to 1.6T.** Nvidia's networking roadmap (see Fig 16) and evolving network architectures point to increasing bandwidth requirements. Large-scale AI training drives substantial GPU-to-GPU traffic, while reasoning and Agentic AI workloads further increase data movement across compute, memory and storage. As AI clusters scale, demand for higher-speed optical connectivity should continue to rise.
- Speed migration drives ASP upside.** The transition from 800G to 1.6T increases lane speeds from 100G to 200G, raising technical requirements across DSPs, drivers, TIAs, lasers, photodetectors and packaging. We expect this increase in complexity to support higher ASPs for critical components. A 200G PD must deliver higher bandwidth while maintaining responsivity, low capacitance and low noise, requiring more advanced epitaxy, device design and process control. We estimate PD ASPs could increase from about US\$0.3 to more than US\$2 as 200G products ramp (see Fig 17). We believe a limited supplier base - Broadcom (Winsemi), GCS and Macom - and GCS's position as the leading 100G PD supplier support its participation in this upgrade cycle.
- Overflow demand supports additional PD opportunities.** Demand for InP-based optical components continues to exceed available supply. We believe constrained industry capacity is driving leading laser makers to quit their own PD production due to ASP differences (Fig 17), and transceiver module makers start to qualify additional suppliers, creating opportunities for independent III-V foundries. As one of the few suppliers to possess InP manufacturing processes, GCS is well positioned to capture incremental wafer-fab and device demand. As 200G PDs enter volume production, we expect GCS to benefit from both higher shipment volumes and a more favourable product mix.
- Substrate supply concerns appear overstated:** Investors often question the availability of InP substrates, a key production input. We expect leading overseas InP substrate suppliers to add capacity (Fig 18), which should gradually improve supply conditions. GCS also appears to have secured sufficient substrate supply for 2H26. Rather than facing raw material constraints, the key challenge may be how quickly GCS can expand manufacturing capacity to meet customer demand.

Figure 15 - Nvidia speed migration

	GPU	Scale-up (In-rack)	Scale-out (Between-rack)	
H100 (2022)	Chip	NV Switch 4 (900 GB/s)	CX7 – 400G	Infiniband / Spectrum4 51.2T
	Interconnect	PCB trace	Layer 1 Transceiver 400G VCSEL	Layer 2/3 Transceiver 800G EML, SiPh
GB200 (2024)	Chip	NV Switch 5 (1.8TB/s)	CX7 – 400G	Infiniband / Spectrum4 51.2T
	Interconnect	DAC (Copper)	Layer 1 Transceiver 400G VCSEL	Layer 2/3 Transceiver 800G EML, SiPh
GB300 (2025)	Chip	NV Switch 6 (3.6TB/s)	CX8 – 800G	Infiniband / Spectrum4 51.2T
	Interconnect	DAC (Copper)	Layer 1 Transceiver 800G EML, SiPh	Layer 2/3 Transceiver 800G EML, SiPh
Rubin (2026)	Chip	NV Switch 7 (3.6TB/s)	CX9 – 800G *2	Infiniband / Spectrum 6 102.4T
	Interconnect	DAC (Copper)	Layer 1 Transceiver 1.6T EML, SiPh / CPO?	Layer 2/3 Transceiver 1.6T EML, SiPh? CPO?
Rubin Ultra (2028)	Chip	NV Switch 8?	CX10?	Infiniband / Spectrum 7 204T
	Interconnect	PCB Backplane? DAC? AEC? NPO? CPO?	CPO?	Transceiver? CPO?

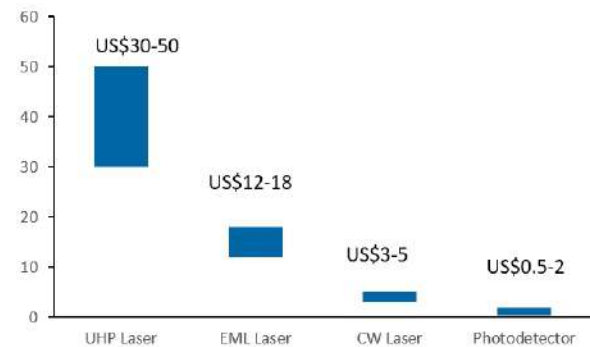
Source: Nvidia, Macquarie Research, Aug 2026

Figure 16 - ASP difference: 100G to 200G PD migration



Source: Macquarie Research, Aug 2026

Figure 17 - Optical chip ASP difference



Source: Macquarie Research, Aug 2026

Figure 18 - Announcement from key suppliers of InP substrate

	Sumitomo Electric	JX Metals	AXT
Country	Japan	Japan	Capacity in China; Listed in US
Investment amount	JPY 17.7bn	JPY 120bn	Capex US\$30-40mn in 2025, US\$100mn in 2027
Recent announcement	Expand capacity 3.1x by 2028, compared to 2024 base.	Expand 10x throughout FY2030	4x of 4Q25 capacity in 2027
Key partnership	LTA with LITE, COHR, Landmark, etc.	LTA with LITE, COHR, VPEC, etc.	LTA with COHR, TW suppliers, etc.

Source: Companies, Macquarie Research, Aug 2026

Shared capacity - GCS's hidden edge

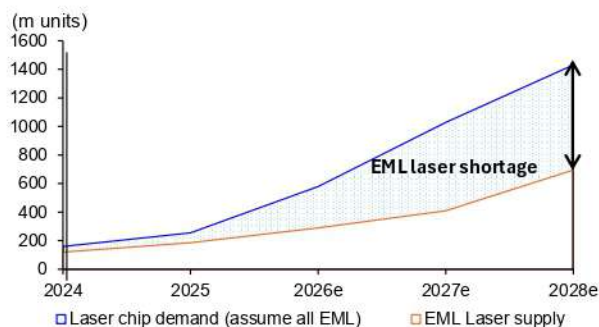
Laser migration creates incremental PD opportunities: Our analysis suggests that InP optical chips share a largely similar manufacturing base (Figure 20). As laser ASPs significantly exceed PD ASPs, leading suppliers continue to allocate more capacity to EML and CW lasers. This shift reduces PD supply and creates incremental opportunities for specialist PD providers.

GCS occupies a differentiated position as the dominant photodetector supplier in the AI optical ecosystem. As 800G deployments accelerate and 1.6T adoption approaches, tight laser supply has prompted several leading laser vendors to reduce or exit PD production. As a result, GCS has emerged as an important supplementary PD capacity provider, given that transceiver modules require both lasers and photodetectors.

Rather than competing directly in the laser market, GCS leverages its existing InP manufacturing platform and established customer relationships to expand its role as a key InP foundry partner. If strong demand persists, we believe third-tier suppliers could gain greater opportunities to enter the laser supply chain, further supporting demand for outsourced InP manufacturing capacity.

GCS is one of the few manufacturers capable of integrating the fragmented III-V compound semiconductor production process, from gratings and foundry services to backend packaging. Unlike CMOS-based silicon semiconductors, III-V compounds (GaN, GaAs and InP) require specialised manufacturing steps. GCS's ability to integrate these processes and maintain stable yields underpins its competitive advantage. We outline the EML/DFB-CW laser diode production process below (Figure 21).

Figure 19 - EML laser shortage



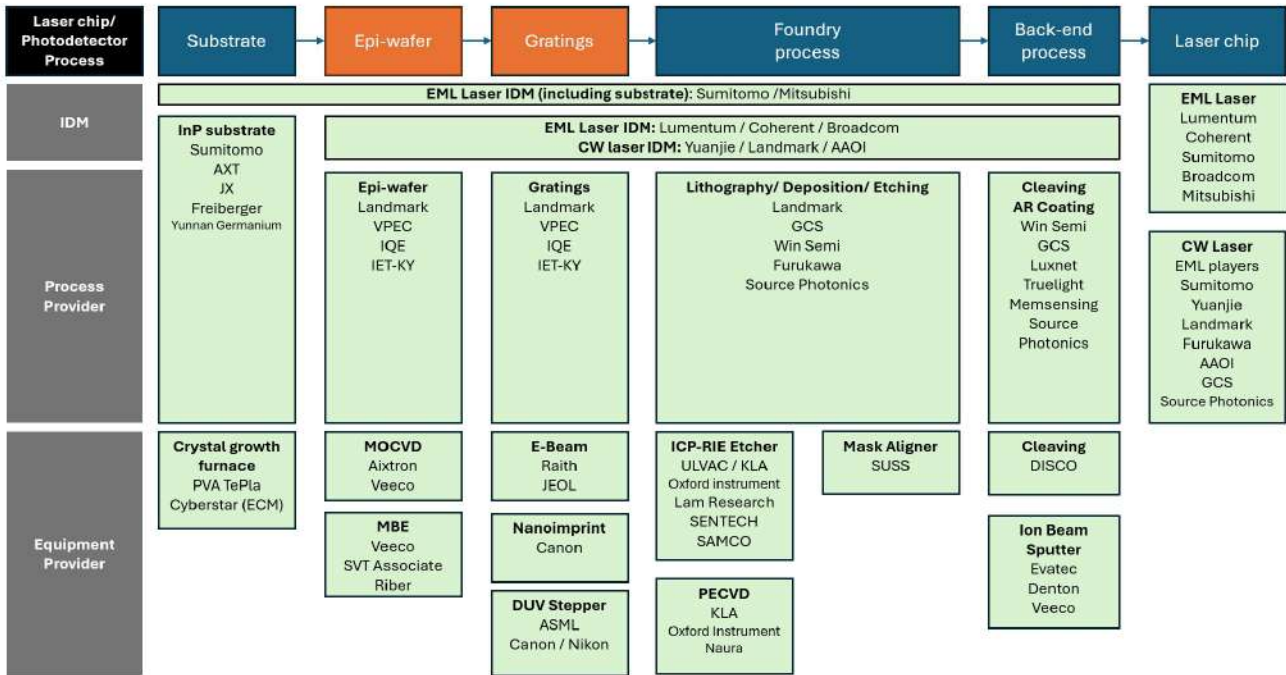
Source: Macquarie Research, Aug 2026

Figure 20 - Equipment needed for Laser / PD

Equipment	Laser/PD
MOCVD	Shared
Lithography	Shared
ICP Etch	Shared
PECVD	Shared
Metallization	Shared
Probe Test	Shared
Cleaving	LD only
Facet Coating	LD only
Responsivity test	PD only

Source: Macquarie Research, Aug 2026

Figure 21 - Laser process overview



Source: Macquarie Research, Aug 2026

Figure 22 - Laser Industry overview

Tier	Company	Capacity plan	Laser product	MQ comment
Tier1- Industry Leading	Lumentum (LITE.US)	>50% YoY EML capacity increase by CY26; new 6-inch InP fab in North Carolina	EML: Commercial 100G/lane, 200G/lane EML; demonstrated 400G/lane EML; CW: 70mW, 100mW, 400mW products; 1st ELSFP for CPO, demo 800mW CW laser	Largest and industry laser maker. Capacity expansion plan seems conservative. We expect further outsourcing of its optical chips.
	Coherent (COHR.US)	Double InP production by CY26 and >2x again by CY27; major 6-inch InP transition	EML: Commercial 100G/lane and 200G/lane EMLs; CW: 70mW, 100mW, 300mW+ UHP laser qualified	Module capacity > laser capacity, still require external supplies of laser, even after its aggressive expansion. Also sees outsourcing opportunity for optical chips.
	Broadcom (AVGO.US)	EML capacity to grow from ~25% in 2026. CW laser capacity is doubling in 2026.	EML: Commercial 100G/lane, 200G/lane CW: 70mW, 100mW, UHP laser ready	Already outsourcing its PD, possibly also CW laser in the future, given its positioning as fabless.
	Sumitomo Electric (5802.JP)	Intra-DC optical device capacity to reach 3.1x 2024 level by 2028	EML: 100G/lane, 200G/lane EMLs; CW: 350mW+ CW lasers in development; 200mW CW laser in production	The key CW laser supplier with largest substrate supply.
Tier2- Production Ready	Yuanjie Semiconductor (688498.CH)	Announced RMB4.27bn expansion for 400G/800G/1.6T applications	EML: Commercial EML, but behind global leaders in 100G/200G deployment CW: 70mW CW lasers; expanding to higher power	Emerging Chinese competitor, stronger in CW lasers than high-speed EMLs
	AAOI (AAOI.US)	Targeting 350% increase in internal laser capacity by 2027. 10x in the next few years	EML: 100G/lane EML CW: 70-100mW CW lasers; demonstrated 400mW external laser source	Primarily captive capacity supporting module business
	Furukawa Electric (5801.US)	>5x laser-chip capacity by 2028; new Japan facility	CW: 70mW, 100mW, 400mW in shipment; 200mW in development	Expansion largely focused on CW/CPO opportunity and later-stage ramp; sourcing Epi from VPEC
	Landmark (3081.TT)	Announced NT\$6bn capacity. Expect capacity double in 2026, 150% YoY in 2027	CW: 50mW, 70mW, multiple wavelength EML: 50G/lane EML	Leading supplier in CW laser market with multiple wavelength capacity; currently in 3"; direct beneficiary from laser shortage.
Tier3- In Development	WIN Semi (3105.US)	Raise 2026 capex budget to NT\$2bn, above 2024.25 level	In cooperation with Siivers	Currently mostly PD product; sourcing Epi from VPEC
	GCS (4991.TT)	~5,500 InP wafers/month with ability to reallocate capacity from RF to optics; optical capacity expansion underway	Developing CW laser capability with VPEC partnership	Best-positioned PD beneficiary, as leading laser makers are occupied with LD orders; sourcing Epi from VPEC

Source: Macquarie Research, Aug 2026

Financial overview

GCS's financial profile has improved following a challenging CY20-24 period. The company is transitioning from losses to sustainable profitability, supported by a stronger product mix, rising AI-driven demand and improving operating leverage.

AI data-centre demand has driven both volume growth and product-mix improvement. Own-brand AOC products increased from 62% of revenue in 2024 to 69% in 2025, and we forecast they will reach 91% by 2028E. Higher-value products and operating leverage drove a sharp recovery in gross margins and earnings. Operating profit reached TW\$404mn in 2025, and we forecast OPM expansion from 18.4% in 2025 to 27.3% in 2028E.

Reported net profit reached only TW\$17mn (EPS TW\$0.15) in 2025, primarily due to TW \$422mn of equity-method losses from non-core investments, partly offset by TW\$50mn of income from Shanghai Zhoujia. The gap between operating profit and net profit suggests that legacy investments, rather than the underlying business, remained the primary drag on earnings.

Investment view

We believe GCS has entered a new phase of development. The company has moved beyond a period characterised by losses, underutilised capacity and earnings drag from legacy investments. We expect AI-related demand, a richer product mix and improving operating leverage to drive the next stage of growth.

We forecast multiple years of strong revenue growth, supported by increasing optical content in AI infrastructure and continued expansion of higher-margin products. We also

expect profitability to improve as scale benefits emerge and non-core earnings headwinds diminish. Our forecasts assume GCS can sustain strong growth while progressively improving margins and profitability over the medium term.

Figure 23 - Income Statement

NT\$m	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Revenues	715	770	1,110	1,556	1,738	1,884	2,124	2,406	1,750	2,195	4,151	8,151	12,711
Gross Profit	390	333	516	739	845	919	1,079	1,228	663	1,035	1,977	4,070	6,776
Operating Profit	182	98	216	318	384	420	516	590	193	404	815	1,910	3,471
PBT	246	134	251	354	420	455	551	626	-235	13	985	2,052	3,613
Net profit	196	110	189	263	311	337	407	462	-237	17	758	1,518	2,657
EPS (NT\$)	1.67	0.92	1.57	2.20	2.60	2.82	3.40	3.86	-2.14	0.15	6.36	12.68	22.20
Profitability													
Gross Margin	54.5%	43.3%	46.4%	47.5%	48.6%	48.8%	50.8%	51.0%	37.9%	47.1%	47.6%	49.9%	53.3%
EBIT Margin	25.5%	12.7%	19.4%	20.5%	22.1%	22.3%	24.3%	24.5%	11.0%	18.4%	19.6%	23.4%	27.3%
Pretax Margin	34.4%	17.3%	22.6%	22.7%	24.2%	24.2%	26.0%	26.0%	-13.4%	0.6%	23.7%	25.2%	28.4%
Net Margin	27.4%	14.3%	17.0%	16.9%	17.9%	17.9%	19.2%	19.2%	-13.5%	0.8%	18.3%	18.6%	20.9%
QoQ Growth													
Sales	3.8%	7.8%	44.2%	40.1%	11.7%	8.4%	12.8%	13.3%					
Operating Profit	25.1%	-46.2%	120.2%	47.5%	20.6%	9.3%	22.8%	14.5%					
Pretax Profit	246.4%	-45.7%	88.2%	40.8%	18.6%	8.5%	21.0%	13.6%					
Net Profit	189.5%	-43.8%	71.2%	39.7%	18.2%	8.4%	20.7%	13.4%					
EPS	177.6%	-44.8%	71.2%	39.7%	18.2%	8.4%	20.7%	13.4%					
YoY Growth													
Sales	51.0%	57.2%	104.3%	126.0%	143.1%	144.6%	91.3%	54.6%	29.6%	25.4%	89.1%	96.4%	55.9%
Operating Profit	392.2%	-16.3%	107.2%	118.7%	110.9%	328.2%	138.8%	85.4%	-184.9%	109.5%	101.7%	134.5%	81.7%
Pretax Profit	na	2179.1%	1520.2%	398.8%	70.7%	240.9%	119.2%	76.8%	na	na	7261.9%	108.4%	76.0%
Net Profit	na	632.6%	1169.3%	281.5%	56.2%	218.0%	119.2%	76.8%	na	na	4240.2%	100.2%	75.0%
EPS	na	627.8%	1130.7%	266.5%	56.1%	206.3%	116.1%	75.4%	na	na	4113.4%	99.3%	75.0%

Source: Company, Macquarie Research, Aug 2026

Figure 24 - Balance Sheet

NT\$m	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & Cash Equivalents	1,106	1,840	442	345	577	1,210	874	948	1,306
Accounts Receivable	207	178	212	245	340	365	824	1,034	1,879
Inventories	362	280	338	258	307	458	1,161	1,674	2,480
Other Current Assets	238	92	101	189	214	246	475	475	475
TOTAL CURRENT ASSETS	1,913	2,390	1,094	1,037	1,438	2,279	3,335	4,131	6,141
Long-Term Investments	1,014	1,691	2,447	1,626	756	528	255	255	255
Gross Fixed Assets	1,936	1,813	2,069	2,268	2,532	2,630	3,106	3,706	4,306
Depreciation	(1,247)	(1,274)	(1,460)	(1,667)	(1,808)	(1,937)	(2,171)	(2,512)	(2,912)
Construction in Progress	7	36	87	129	376	61	41	41	41
Net Fixed Assets	696	575	696	730	1,099	754	976	1,235	1,434
Other L/T Assets	275	246	281	197	213	285	452	469	486
TOTAL ASSETS	3,898	4,902	4,517	3,590	3,506	3,846	5,017	6,089	8,316
Short-Term Debt	20	20	20	20	20	77	86	86	86
Accounts Payable	22	2	8	40	39	28	70	101	149
Current Portion of Long-Term Debt	23	10	43	329	7	7	7	7	7
Other Current Liabilities	135	123	334	127	137	230	209	209	209
TOTAL CURRENT LIABILITIES	200	155	406	516	203	341	372	403	451
Long-Term Debt (ex. current portion)	107	189	329	0	164	150	151	151	151
Other Long Term Liabilities	64	59	82	90	118	110	130	130	130
TOTAL LIABILITIES	371	403	816	606	484	601	654	684	733
Common Stocks	914	1,107	1,108	1,113	1,123	1,139	1,209	1,209	1,209
Reserves	1,710	2,944	2,832	2,680	1,959	2,013	2,916	2,916	2,916
Retained Earnings	1,145	715	(230)	(792)	(237)	17	313	1,354	3,533
FX and Other Adjustments	(151)	(254)	(10)	(16)	176	75	(75)	(75)	(75)
Treasury Stock	(91)	(13)	0	0	0	0	0	0	0
Minority Interest	0	0	0	0	0	0	0	0	0
SHAREHOLDERS' EQUITY	3,527	4,499	3,701	2,985	3,021	3,244	4,363	5,405	7,583
TOTAL LIABILITIES AND EQUITY	3,898	4,902	4,517	3,590	3,506	3,846	5,017	6,089	8,316

Source: Company data, Macquarie Research, Aug 2026

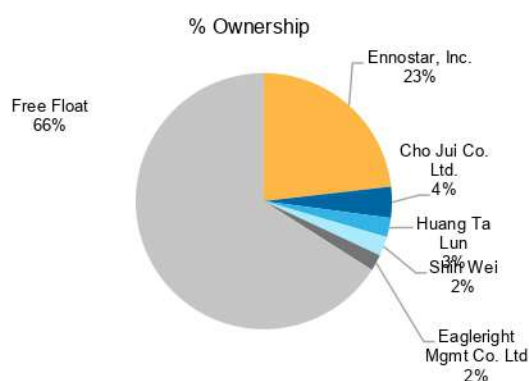
Figure 25 - Cash flow statement

NT\$m	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Net Income	(114)	(378)	(940)	(792)	(237)	17	743	1,498	2,637
Depreciation	119	112	125	136	141	129	234	341	401
Amortisation	7	6	7	7	4	2	3	3	3
Investment (Gain)/Losses - Equity	252	461	734	757	767	372	82	0	0
(Gain)/Losses from Disposals	2	25	(2)	257	218	13	162	0	0
Incr/(Decr) in Provisions	0	0	0	0	0	0	0	0	0
Net Change in Working Capital	117	100	(95)	72	(143)	(105)	(934)	(691)	(1,604)
Other	(9)	(111)	105	(504)	(307)	(32)	(193)	0	0
Cash from Operations	374	216	(67)	(67)	444	396	97	1,151	1,437
Capital Expenditures	(70)	(133)	(183)	(93)	(448)	(59)	(491)	(600)	(600)
Acquisition of L/T Investments	0	0	0	0	0	0	0	0	0
Proceeds from Disposal of S/T Investments	0	0	0	0	0	0	0	0	0
Other	(1,019)	(321)	(1,397)	114	381	206	(312)	0	0
Cash from Investing	(1,089)	(454)	(1,581)	22	(67)	147	(802)	(600)	(600)
Net Change in LT Debt	93	204	176	0	0	0	0	0	0
Net Change in ST Debt	0	0	0	0	0	56	8	0	0
Dividends Paid	(46)	0	0	0	0	0	(463)	(477)	(479)
Proceeds from Share Issuance	0	816	0	0	0	0	786	0	0
Employees & Director Bonus	0	0	0	0	0	0	0	0	0
Other Financing Cash Flows	(25)	(47)	(32)	(53)	(159)	51	3	0	0
Cash from Financing	22	973	144	(53)	(159)	107	335	(477)	(479)
Currency Adjustments	(40)	(2)	107	1	13	(16)	35	0	0
Net Cash Flow	(733)	733	(1,398)	(97)	231	634	(336)	74	358
Free Cash Flow	304	83	(250)	(159)	(4)	337	(394)	551	837

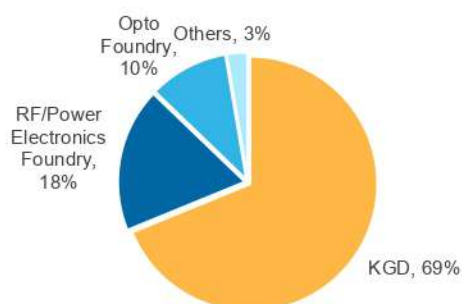
Source: Company data, Macquarie Research, Aug 2026

Company overview

- **GCS Holdings (4991 TT)** is a pure-play compound semiconductor foundry and optical chip manufacturer serving RF and optoelectronics markets. The company provides III-V compound semiconductor wafer fabrication services across GaAs, InP and GaN technologies and sells proprietary optoelectronic products, including photodetectors, to customers in wireless, telecoms and power electronics markets.
- GCS has established a position in compound semiconductors through its manufacturing expertise in III-V materials and its exposure to optical communications applications. In our view, this positions the company to benefit from growing demand for compound semiconductor devices across communications and optical networking markets.

Figure 26 - Ownership structure

Source: FactSet, Macquarie Research Aug 2026

Figure 27 - 2025 product mix

Source: Company data, Macquarie Research, Aug 2026

Key risks

- **Delayed 1.6T transceiver deployment.** A delay in industry-wide 1.6T transceiver adoption could slow GCS's revenue growth and earnings expansion. Slower customer qualification, extended inventory digestion or delays in data centre deployments could postpone volume shipments of next-generation transceivers and reduce demand for GCS's optical chip products. Lower shipment volumes could weaken utilisation rates and operating leverage. Given our investment thesis is linked to the 800G-to-1.6T upgrade cycle, a material delay could result in earnings downgrades and valuation pressure.
- **Delayed 200G PD development.** GCS's growth outlook depends in part on the successful development, qualification and commercialisation of its next-generation 200G photodetectors. Delays in product development, customer qualification or mass production could postpone content gains and limit the company's ability to capture demand from higher-speed optical modules. Customers may also retain existing solutions for longer or adopt competing products. As a result, the expected increase in ASPs and revenue contribution from 200G PDs could materialise later than forecast, weighing on revenue growth, margins and earnings.
- **Failure to qualify CW lasers.** GCS's long-term growth outlook partly depends on successfully qualifying and commercialising its CW laser products. Customer qualification in the optical industry typically requires extensive performance and reliability testing before volume production can begin. Failure to meet customer requirements could limit GCS's ability to expand into higher-value optical components and increase content per module. This could reduce future revenue opportunities, slow customer diversification and weaken the company's competitive positioning in the laser market.
- **Faster-than-expected adoption of SiPh/CPO:** A faster-than-expected transition toward silicon photonics (SiPh) and co-packaged optics (CPO) architectures could adversely affect demand for some of GCS's existing optical chip products. As data-centre operators and networking equipment vendors seek higher bandwidth and lower power consumption, adoption of alternative optical technologies may accelerate beyond current expectations. If GCS is unable to adapt its product portfolio, develop competitive solutions, or secure a meaningful position within emerging SiPh and CPO ecosystems, its market share and long-term growth prospects could come under pressure. This may result in lower-than-expected revenue contribution from legacy products and increase execution risk during the technology transition.

Key Quant Findings

The quant model currently holds a strong negative view on GCS Holdings. The strongest style exposure is Growth, indicating this stock has good historic and/or forecast growth. Growth metrics focus on both top and bottom line items. The weakest style exposure is Profitability, indicating this stock is not efficiently converting investments to earnings; proxied by ratios like ROE or ROA.

Macquarie Alpha Model: Key rankings

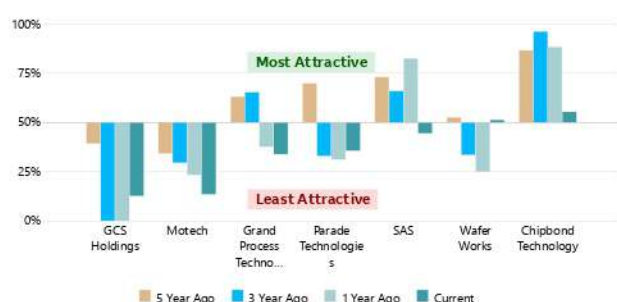
The Macquarie Quant's flagship Alpha model is a dynamic multi-factor model based on a staple of quant factors such as value, momentum, revisions, quality, and risk.

	Global	Market (Country)	Sector
	Whole Universe	Taiwan	Semiconductors & Semiconductor Equip.
Macquarie Alpha Model	15633/17970	854/977	570/670
Fundamental (Consensus) *	4838/17970	138/977	213/670

* based on Total Shareholder Return = Consensus Price target / Current Price

Current and Historical Alpha Model Rank

The chart shows the Macquarie Alpha model market ranking against the company's peers and over recent history.



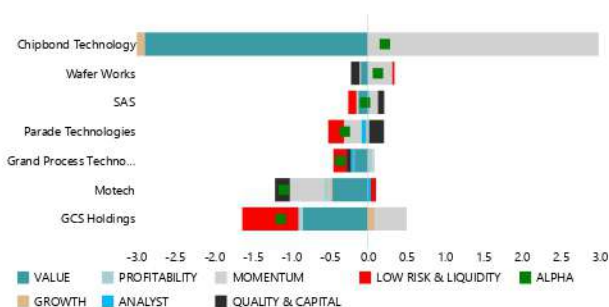
Alpha Model Decomposition

The Macquarie Alpha is decomposed into its sector and market relative factor & styles exposures (a higher/better percentile is coded in green, whilst lower in red).

Factors / Styles	Percentile relative to		Core factors in definition
	sectors (/670)	market (/977)	
ALPHA	15%	13%	Built from the styles below
VALUE	20%	1%	Book, CF, Yield, Earnings Multiples
ANALYST	60%	87%	Revisions (Earnings, Recommendations)
MOMENTUM	56%	83%	Price Momentum
GROWTH	96%	99%	EPS, Sales (Forecast, Historic)
PROFITABILITY	13%	19%	ROE, Margin, Asset Turnover
QUALITY	65%	75%	Accruals, Earn Stability, Cash Conversion
CAPITAL	75%	36%	Investment/Capex, Net share issuance
LIQUIDITY	30%	30%	Size, Turnover, Analyst Coverage
LOW RISK	3%	1%	Beta, Volatility, Earn.Cert, Leverage
TECHNICAL	47%	69%	MACD, RSI, Bollinger, Williams R, etc

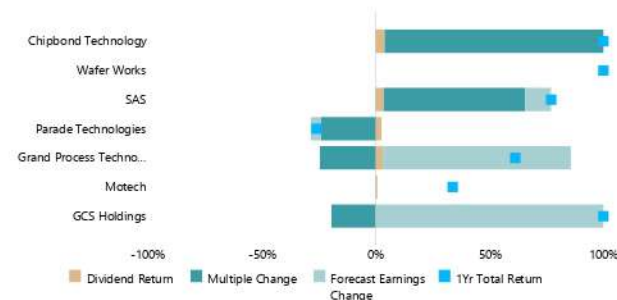
Factors driving the Alpha Model vs peers

For the comparable firms this chart shows the key underlying styles and their contribution to the current overall raw Alpha score.



Drivers of Stock Return

Breakdown of 1-year total return (local currency) into returns from dividends, changes in forward earnings estimates and the resulting change in earnings multiple.



Macquarie Style Returns over last year

Recent performance to Macquarie style factors

	Monthly Factor Long-Short Return for												Last 5 Years (ann)	Last 10 Years (ann)
	Jul-25	Jun-25	May-25	Apr-25	Mar-25	Feb-25	Jan-25	Dec-24	Nov-24	Oct-24	Sep-24	Aug-24		
ALPHA	10%	1%	3%	3%	4%	0%	0%	3%	3%	2%	-3%	-6%	9%	6%
VALUE	19%	-9%	1%	-7%	6%	0%	-2%	3%	5%	4%	-4%	-4%	14%	5%
ANALYST	-12%	7%	4%	12%	-2%	0%	6%	5%	-2%	1%	4%	1%	12%	13%
MOMENTUM	-33%	15%	17%	19%	-4%	3%	8%	6%	-1%	-2%	0%	-6%	-4%	2%
GROWTH	-19%	9%	7%	7%	-4%	-1%	4%	2%	-2%	0%	5%	8%	2%	1%
PROFITABILITY	3%	6%	1%	-4%	3%	0%	-4%	1%	3%	0%	-4%	-2%	-3%	-2%
QUALITY	7%	-3%	-2%	-4%	2%	-1%	-4%	-1%	1%	-4%	-6%	1%	-5%	-2%
CAPITAL	14%	-7%	-1%	-12%	2%	-2%	2%	-1%	3%	-3%	-1%	-2%	3%	-4%
LIQUIDITY	-11%	8%	10%	7%	-1%	3%	1%	2%	-1%	-1%	3%	3%	-3%	3%
LOW RISK	29%	-6%	-8%	-13%	6%	-2%	-3%	-2%	5%	1%	-6%	-11%	-5%	-6%

Source (all charts): FactSet, Refinitiv, and Macquarie Quant. For more details on the Macquarie Alpha model or for more customised analysis and screens, please contact the Macquarie Global Quantitative Team: maccaperequitiesresearchquantglobal@macquarie.com. Explanation for items on this page can be found at <https://www.macquarieinsights.com/rp/d/r/p/OTUyMzg1>

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Important Disclosures

Recommendation definitions	Volatility index definition	Financial definitions
Macquarie – Asia and USA Outperform – expected return >10% Neutral – expected return from -10% to +10% Underperform – expected return <-10% Macquarie – Australia/New Zealand Outperform – expected return >10% Neutral – expected return from 0% to 10% Underperform – expected return <0% During periods of share price volatility, recommendations and target prices may occasionally and temporarily be inconsistent with the above definitions. Recommendations – 12 months 12-month target – Expected share price in 12 months Valuation – The company's estimated fair value share price based on the disclosed valuation methodology Note: Quant recommendations may differ from Fundamental Analyst recommendations	This is calculated from the volatility of historical price movements. Very high – highest risk – Stock should be expected to move up or down 60–100% in a year – investors should be aware this stock is highly speculative. High – stock should be expected to move up or down at least 40–60% in a year – investors should be aware this stock could be speculative. Medium – stock should be expected to move up or down at least 25–40% in a year. Low – stock should be expected to move up or down at least 15–25% in a year. * Applicable to select stocks in Asia/Australia/NZ Note: expected return is reflective of a Medium Volatility stock and should be assumed to adjust proportionately with volatility risk	All "Adjusted" data items have had the following adjustments made: Added back: goodwill amortisation, provision for catastrophe reserves, IFRS derivatives & hedging, IFRS impairments & IFRS interest expense Excluded: non recurring items, asset revals, property revals, appraisal value uplift, preference dividends & minority interests EPS = adjusted net profit / efpowa* ROA = adjusted ebit / average total assets ROA Banks/Insurance = adjusted net profit / average total assets ROE = adjusted net profit / average shareholders funds Gross cashflow = adjusted net profit + depreciation *equivalent fully paid ordinary weighted average number of shares All Reported numbers for Australian/NZ listed stocks are modelled under IFRS (International Financial Reporting Standards).

Recommendation proportions for quarter ending 30 Jun 2026

	AU/NZ	Asia	USA	
Outperform	65.38%	70.02%	64.29%	(for global coverage by Macquarie, 1.91% of stocks followed are investment banking clients)
Neutral	27.62%	16.78%	35.71%	(for global coverage by Macquarie, 0.40% of stocks followed are investment banking clients)
Underperform	6.99%	13.19%	0.00%	(for global coverage by Macquarie, 0.00% of stocks followed are investment banking clients)

Company-Specific Disclosures

Company Name	Disclosure
GCS Holdings (4991 TT) Outperform 12-month target: TWD710.00 - PER Valuation: TWD 710.00 - PER Price: TWD445.00	None
Visual Photonics Epitaxy (2455 TT) Outperform 12-month target: TWD600.00 - PER Valuation: TWD 600.00 - PER Price: TWD428.00	None
LandMark Optoelectronics (3081 TT) Outperform 12-month target: TWD3,345.00 - PER Valuation: TWD 3,345.00 - PER Price: TWD3,255	None

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Recommendation history

Company name

GCS Holdings (4991 TT)



LandMark Optoelectronics (3081 TT)



13-Aug-2026	Outperform	TWD 3345.00
22-Apr-2026	Outperform	TWD 2714.93
29-Jan-2026	Outperform	TWD 950.43
13-Nov-2025	Outperform	TWD 396.70

Visual Photonics Epitaxy (2455 TT)



13-Jul-2026	Outperform	TWD 600.00
28-Apr-2026	Outperform	TWD 500.00
2-Mar-2026	Outperform	TWD 300.00
13-Nov-2025	Outperform	TWD 236.00

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